

Qi Xu

PhD Student in Artificial Intelligence @ Westlake University

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EDUCATION



Westlake University

PhD Student in Artificial Intelligence

Supervised by [Peidong Liu](#); Research: Embodied AI, Multimodal Large Language Models

2026 – Present



Wuhan University

Master in Pattern Recognition and Intelligent Systems

Research: Computer Vision, 3D Reconstruction, Multimodal Learning

2023 – 2026



Wuhan University

B.S. in Spatial Information and Digital Technology

2019 – 2023

EXPERIENCE



ByteDance, TikTok Live – LLM Applications Team

2025 – 2026

Research Intern

- Developed and optimized **Multi-Modal Large Language Models (MLLMs)** for temporal grounding tasks using SFT and Reinforcement Learning (RL) pipelines.
- Built scalable data pipelines for massive MLLM training, enabling efficient collection and annotation of high-quality temporal grounding datasets.
- Conducted **large-scale distributed training on hundreds of GPUs**, optimizing training efficiency and model convergence for production-level deployment.
- Designed agentic workflows leveraging LLMs to automate real-time interaction analysis and optimize content distribution within the TikTok Live ecosystem.

SELECTED PUBLICATIONS



SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment

[NeurIPS 25 Spotlight](#) [Project Page](#) [ArXiv](#) [Code](#)

Qi Xu*, Dongxu Wei*, Lingzhe Zhao, Wenpu Li, Zhangchi Huang, Shunping Ji, Peidong Liu

2025

Proposed an alignment-free framework achieving SOTA 3D reconstruction and scene understanding from unposed images via shared pixel-aligned 3D representation.



Towards One-to-Many Temporal Grounding

[ICML 26](#) [Project Page](#) [ArXiv](#)

Qi Xu*, Yue Tan*, Shihao Chen, Jiahao Meng, Anran Wang, Shunping Ji, Hao Fei, Xiangtai Li

2026

Introduced the first systematic solution for OMTG, featuring a 56k-sample dataset and RL training with CoT-based rewards; outperformed Gemini 2.5 Pro by 15.85%.



Any 3D Scene is Worth 1K Tokens: 3D-Grounded Representation for Scene Generation at Scale

[NeurIPS Submission](#) [Project Page](#) [ArXiv](#) [Code](#)

Dongxu Wei*, **Qi Xu***, Zhiqi Li, Hangning Zhou, Cong Qiu, Hailong Qin, Mu Yang, Zhaopeng Cui, Peidong Liu

2026

First approach to perform 3D scene generation directly within an implicit 3D latent space using 3D Diffusion Transformers.



HiCI: Hierarchical Construction-Integration for Long-Context Attention

[ICML 26](#) [ArXiv](#) [Code](#)

Xiangyu Zeng, **Qi Xu**, Yunke Wang, Chang Xu

2026

Efficient long-context modeling using hierarchical attention, extending LLaMA-2 to 100K tokens with only 5.5% additional parameters.



Watch, Remember, Reason: Human-View Video Understanding with MLLMs

[arXiv 26](#) [ArXiv](#) [GitHub](#)

Jiahao Meng, Yue Tan, **Qi Xu**, Kuan Gao, Weisong Liu, Yanwei Li, Jason Li, Lingdong Kong, Haochen Wang, Qianyu

2026

Zhou, Jiangning Zhang, Guangliang Cheng, Yunhai Tong, Lu Qi, Minghsuan Yang

Surveyed human-view video understanding with MLLMs through watching, remembering, and reasoning abilities for long, multimodal, and knowledge-intensive video scenarios.



VideoZeroBench: Probing the Limits of Video MLLMs with Spatio-Temporal Evidence Verification

[NeurIPS Submission](#) [Project Page](#) [ArXiv](#) [Code](#)

Jiahao Meng, Tan Yue, **Qi Xu**, Haochen Wang, Zhongwei Ren, Weisong Liu, Yuhao Wang, Renrui Zhang, Yunhai

2026

Tong, Haodong Duan

A hierarchical benchmark exposing the gap between surface-level answer correctness and genuine evidence-based reasoning in long-video understanding.



E-MoFlow: Learning Egomotion and Optical Flow from Event Data via Implicit Regularization

NeurIPS 25 [Project Page](#) [ArXiv](#) [Code](#)

Wenpu Li, Bangyan Liao, Yi Zhou, **Qi Xu**, Pian Wan, Peidong Liu

An unsupervised framework for joint ego-motion and optical flow estimation using implicit neural representations and differential geometric constraints.

2025

PUBLICATIONS WITH ADVISED STUDENTS



SupIR-GS: Thermal Infrared Super-Resolution Novel View Synthesis with Imaging-Calibrated 3DGS

Submission Score: 443 Full Score: 6 Corresponding Author

Jin Liu, Haodong Li, Jiagang Chen, Dabin leng, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, **Qi Xu**[†], Zhiwen Zheng, Xingru Huang[†]

2026

Reconstructs high-resolution 3D thermal scenes from low-resolution inputs via physics-informed degradation modeling.



Physically Grounded Dual-Opacity Gaussian Splatting for Joint RGB-TIR Reconstruction

Submission Score: 444 Full Score: 6 Corresponding Author

Jin Liu, Dabin leng, Jiagang Chen, Haodong Li, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, Zhiwen Zheng, Xingru Huang[†], **Qi Xu**[†]

2026

A unified RGB-TIR reconstruction framework resolving cross-spectral visibility conflicts through thermal field modeling.

TECHNICAL SKILLS

Deep Learning: PyTorch, Transformers, MLLMs, Diffusion Models, Reinforcement Learning

Engineering: Ray, Distributed Training (Multi-GPU/Cluster), Linux, Data Pipelines

Research Interests: Embodied AI, Multimodal Large Language Models, 3D Foundation Models

Languages: English (CET6), Mandarin (Native)